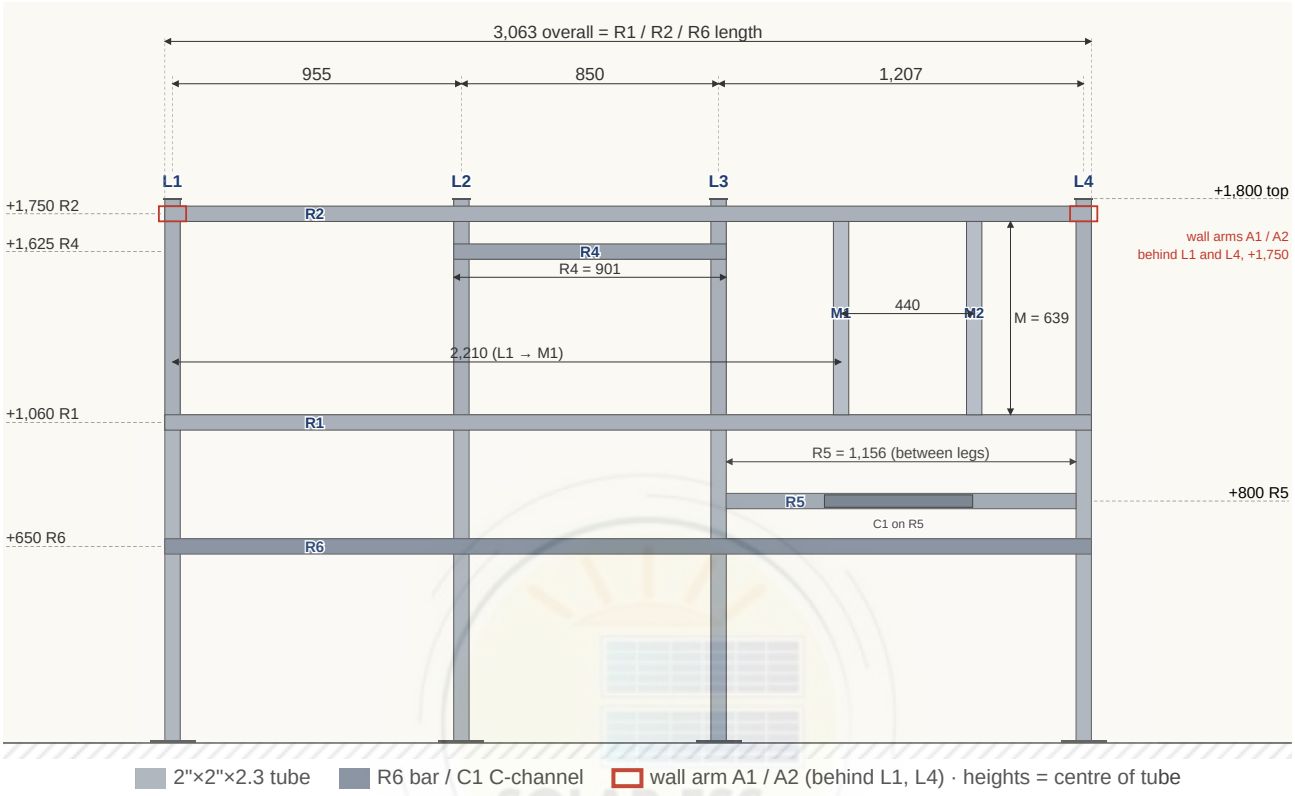
Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694, 089 632 5829
www.solaress.org · www.โซลาร์เอสส์.com

WELDER JOB · STEEL EQUIPMENT FRAME (STR-INV-02)

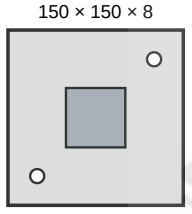
Fabricate and install a galvanized 2"×2"×2.3 steel frame for an inverter, AC box and DC box, standing 500 mm off the wall on 2 wall arms · all dimensions in mm · scale 1:25 on A4

Front view · complete frame

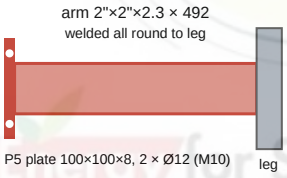
Scale 1:25 0 1 m



Details

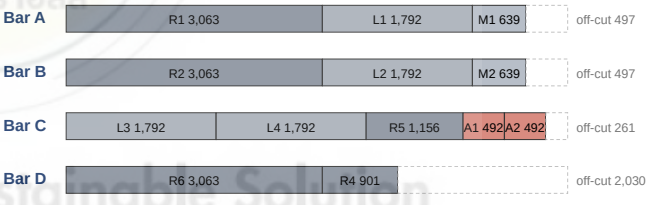


P4 base plate



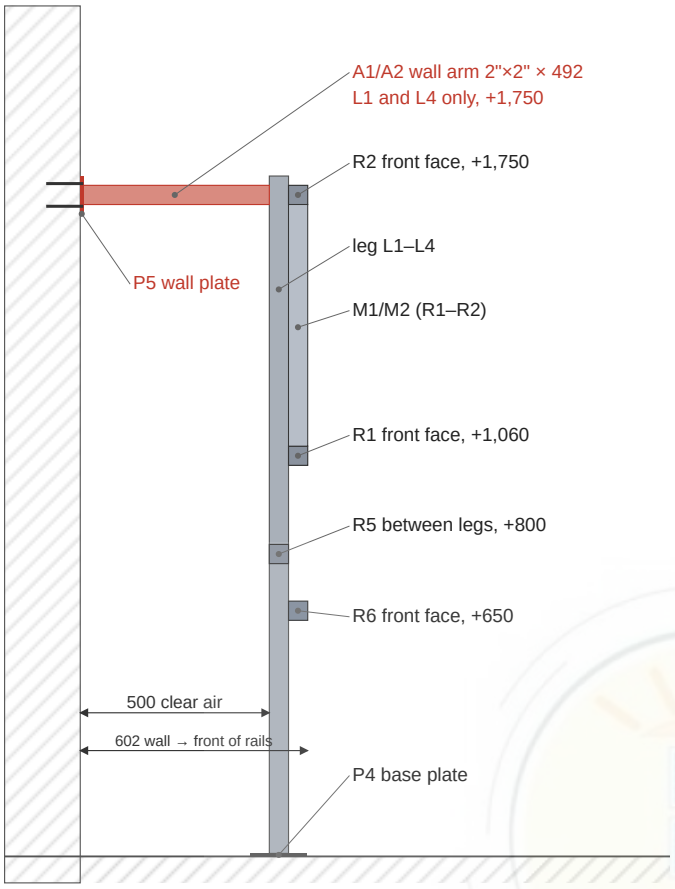
Wall arm + P5 plate (500 clear)

Cut list from 4 × 6 m lengths (3 mm kerf)



Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694, 089 632 5829
www.solaress.org · www.โซลาร์เอสส์.com

Side section (1:20)



Bill of materials

Ref	Item	Size	Qty	Each	Total
Steel tube 2"×2"×2.3 (≈ 3.5 kg/m), order 4 lengths × 6 m					
L1–L4	Legs, cap 60×60×3 on top	1,792 + 8 plate	4	1,792	7.17 m
R1, R2	Rails at 1,060 / 1,750, on front face of legs	3,063	2	3,063	6.13 m
R4	Rail at 1,625, front face L2–L3	901	1	901	0.90 m
R5	Bar at 800, between L3 and L4	1,156	1	1,156	1.16 m
M1, M2	Verticals R1–R2 at 2,210 / 2,650	639	2	639	1.28 m
R6	Bar at 650 (old wireway position), front face of legs	3,063	1	3,063	3.06 m
A1, A2	Wall arms behind L1 and L4 at +1,750, 90° to wall	492	2	492	0.98 m
—	Total tube ≈ 20.7 m ≈ 72 kg				
Angles, plates and fixings					
P4	Base plate, 2 × Ø12 holes	150×150×8	4	1.4 kg	5.7 kg
P5	Wall plate on each arm, 2 × Ø12	100×100×8	2	0.63 kg	1.3 kg
—	Top caps, plate 60×60×3	60×60×3	4	—	4 pcs
C1	C-channel 41×41×2.5 + 2 × M10	490	1	—	0.5 m
—	Wedge anchors M10 × 80	floor 8, wall 4	12	—	12

Scope of work (welder)

1. Check the wall and floor; mark leg centres 955 / 850 / 1,207, legs' back face 500 mm off the wall (air clearance).
2. Cut all parts to the cut list; weld base plates and top caps.
3. Weld R1 and R2 to the front face of the legs, R4 between L2–L3, R5 between L3–L4, M1/M2 between R1 and R2.
4. Weld R6 to the front face of the legs at +650, parallel to R1 and R2 (same length 3,063).
5. Stand the frame, anchor base plates (2 × M10 each), weld the 2 wall arms (492 + 8 mm plate) square to the back of L1 and L4 at +1,750 and anchor each plate with 2 × M10. All 4 base plates must be anchored.
6. Bolt C1 on R5. Drill box and bracket holes only where Solar ESS marks them on site (Ø9 for M8).

Quality and finish · Fillet welds 4 mm all round on every joint (E7018 or E6013), spatter ground off · legs plumb and rails level ± 2 mm, leg spacing ± 3 mm, arms square to the wall and 500 mm gap ± 5 mm · hot-dip galvanize after welding, or zinc-rich primer + 2 coats cold galvanizing on every weld and cut · cap all open tube ends, no sharp edges · welder brings welding set, grinder, drill, consumables and PPE; Solar ESS supplies the steel above unless agreed · about 2–3 days, welder + helper · wireways, conduits, equipment and cabling are by others.

Why 2 arms at L1 and L4, +1,750 · The equipment hangs in front of the frame and tries to tip it forward (≈ 0.8 kN·m). Arms at the top give the longest lever: ≈ 0.2 kN per arm against ≈ 5 kN for an M10 anchor. Arms at both end legs stop the frame twisting; the welded rails carry this to L2 and L3. The arm sits level with R2, so the load passes straight through the leg. The floor anchors on all 4 base plates are part of the hold and must be fitted.